

Iris Flower Classification

Oasis Infobyte Data Science Internship (OIBSIP)

Objective

The objective of this project is to build a Machine Learning model that can classify Iris flowers into three different species based on their measurements.

Dataset

The dataset contains 150 records of Iris flowers with the following features:

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width
- Species

The target variable is **Species**, which includes:

- Iris-setosa
- Iris-versicolor
- Iris-virginica

Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-Learn

Machine Learning Algorithm

Decision Tree Classifier

Project Workflow

1. Import required libraries
2. Load the dataset
3. Data preprocessing
4. Exploratory Data Analysis (EDA)
5. Label Encoding
6. Train-Test Split
7. Model Training
8. Prediction
9. Model Evaluation

Results

The Decision Tree Classifier achieved high accuracy in classifying Iris flower species.

Files Included

- ParveshAhamed_Task1.ipynb
- task1 .csv
- requirements.txt
- README.md

Author

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Internship

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